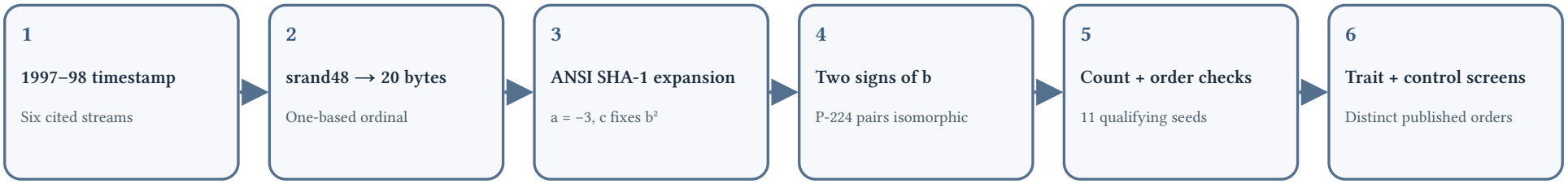
The eleven skipped prime-field seed candidates

Structural assessment · 16 October 2026 · exact data and code in the accompanying study directory

Reconstruction and assessment boundary



Source: Valsorda article; NIST SP 800-186 C.3.1; reproducible audit. Assessed data SHA-256: 370e79ce0bff8f26…

Question and method

Filippo Valsorda's recovered seed streams name eleven unpublished candidates that passed the visible criteria: five at 239 bits, two at 192 bits, one at 256 bits, and three at 224 bits. We reconstructed their complete 160-bit seeds with the 48-bit srand48 recurrence; all ten published prime-field seeds reproduced exactly. NIST's SHA-1 expansion then defines c, and the equations use $a = -3$ with $c\,b^2 = -27$ modulo p. Sage point counting, proven-prime n, an on-curve nonidentity point killed by n, and Hasse-interval uniqueness check each candidate's order. Published p, a, b, G, n and seed are OpenSSL controls. Source details and commands are in report.md, audit.py and postprocess.py.

All eleven have prime order, cofactor 1, n bit length no greater than p bit length, $n \neq p$, j outside 0 and 1728, and embedding degree greater than 100. A complete scan through fundamental CM discriminants of absolute value at most 2,000,000 found none; this excludes noninteger endomorphisms below degree 500,000. Exact CM fields and endomorphism conductors are unresolved.

Candidate-by-candidate traits

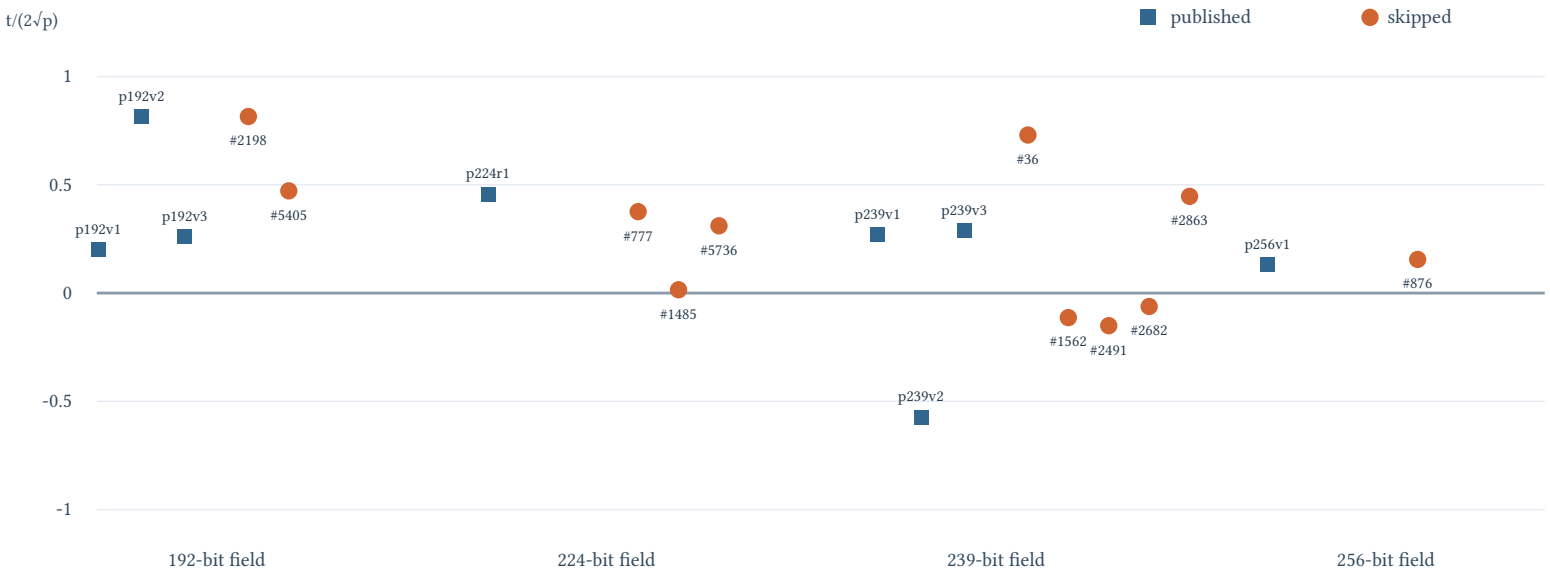
The ordinal is one-based. Trace is $t/(2\sqrt{p})$, $t = p + 1 - n$. Twist factors shown are proven factors at most 10,000. P means the residual twist factor is proven prime and the factorization is complete; C means a composite residual remains. Full p, b, n, generator and EC1 representation identity appear in assessed.json.

Bits	Ordinal	Recovered seed, hex	n vs p	Trace	Twist factors ≤10,000	Rest
239	36	`25e56088185f8c3207b166cd2e007b88276afe78`	<	+0.730	139	C
239	1562	`d09bfb7db85672dc700269f9189983f25f81958d`	>	−0.113	41	C
239	2491	`1256fdd46eeb96b7d312ddb1de99bff159cf6701`	>	−0.151	43, 173, 307, 953	C
239	2682	`d2c18e22558203444a2b33ea97c178675b438484`	>	−0.062	29	C
239	2863	`3a700d3e6bf03255949804ba2a8677550c19ee11`	<	+0.447	none found	C
192	2198	`692d90186057951de011733105bf1b3d3b0fc451`	<	+0.815	11, 127, 5827	P
192	5405	`3ad10ff61797bc7258480cadd4e59c1ac1fb66d1`	<	+0.472	7, 19^2	P
256	876	`50c71a63f911db7b2f31618ecf609d37154437f9`	<	+0.156	3, 11, 37, 8081	C
224	777	`d00720e7691d403449c0d91be84bfb2c7501a267`	<	+0.376	3, 5, 683	C
224	1485	`18cfdbcbb825568171212bc1610a5b8a8004e5c4`	<	+0.015	3, 31	C
224	5736	`3c170a30ee6709df9b4c9ea0b6c1e95fcd6200ce`	<	+0.311	3, 1607, 3371	C

Comparison with the published curves

Frobenius trace across published and skipped curves

Exact derived value t/(2√p), where t = p + 1 − #E(Fp). Eleven skipped seeds; eight same-field published controls.



Data: traits.json and assessed.json (SHA-256 370e79ce0bff8f26…). Label # is the article's one-based seed ordinal. No sampling uncertainty.

All eleven omitted group orders differ from published orders over the same prime field, so there is no base-field isogeny to a published curve in the same field. Three omitted 239-bit curves have n greater than p, but published prime239v2 does too: n < p is not the missing selection rule. Trace signs, small twist factors, and the bounded CM screen reveal no rule that picks precisely the published subset. The three P-224 seeds yield two b signs each; because −1 is a square in that field, each pair is isomorphic by $(x,y) \longrightarrow (-x,u^3y)$, $u^2 = -1$.

The two 192-bit omitted twists have complete factorizations into the table's small factors and one proven large prime. The other nine twist residuals are composite and not completely factored. Twist factors are relevant to applications admitting unchecked off-curve points, not a weakness of the prime-order group itself.

Inference boundary

The historical reason the eleven curves were skipped remains unidentified. This study reconstructed precisely the candidates named in the two primary accounts; it did not enumerate every earlier rejected block or other timestamps. Neither the surviving numbers nor this trait comparison reveal how many separate timestamp runs took place. A historical code or log recovery, or a predeclared deterministic rule that correctly predicts accepted and skipped cases on a held-out run, would resolve the selection question. The current screens do not assess a hypothetical undisclosed map, extension-field transfer, or intermediate-cost advantage; each would need a construction and costed, independently checked subgroup transport.

Sources and reproducibility

Valsorda, The NIST Elliptic Curves Seeds Bounty Has Been Claimed, 16 October 2026 [retrieved]. Anthropic, Claude reveals ECDSA curve origins, 16 October 2026 [retrieved]. **NIST, SP 800-186, Appendix C.3.1 [retrieved].** Local exact records: traits.json, assessed.json, audit.py, postprocess.py and run logs [internal].